How to Identify a Smoker?

An in-depth analysis on nation-wide behavior in relation to tobacco consumption

Smoking is a high-risk habit that puts all parts of the body at risk. Governments should therefore be motivated in taking steps towards minimizing its country's tobacco consumption. In this research we set out to understand the environmental conditions that incentivize smoking behavior. We believe, that this approach could advance governmental attempts at bettering the health of the people.

Our research looks at *tobacco consumption* and *the percentage of the population who are daily smokers* and uses them for a comparative analysis in relation to a variety of lifestyle measures. In our correlational analysis, among other measures, *Vegetable supply* and *Sugar supply* have a negative correlation with the percentage of *Daily smokers, Tobacco, and Alcohol consumption.* However, further research is needed to further prove this phenomenon.

For a more focused evaluation we limited our data to the countries with highest and lowest average percentage of *Daily smokers*. We found that, though the groups do not show a clear pattern, in the high-percentage-countries (Denmark, Netherlands, Russia, Türkiye, and Japan) some measures show a negative correlation coefficients up to 0.6 with the *Daily smokers.*

While our findings are not definitive, we believe this approach should be further explored, introducing more measures and data. Furthermore, while recognizing global trends, it is important to acknowledge and explore the specific environmental conditions that might have a role both in national and regional biases.

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Data Science Project

The main measures used in the project:

The measures were all recorded between the years 1960-2023, either describing the share of population with that quality or an average, per capita measure in the given country.

Daily smokers: share of population who are daily smokers.

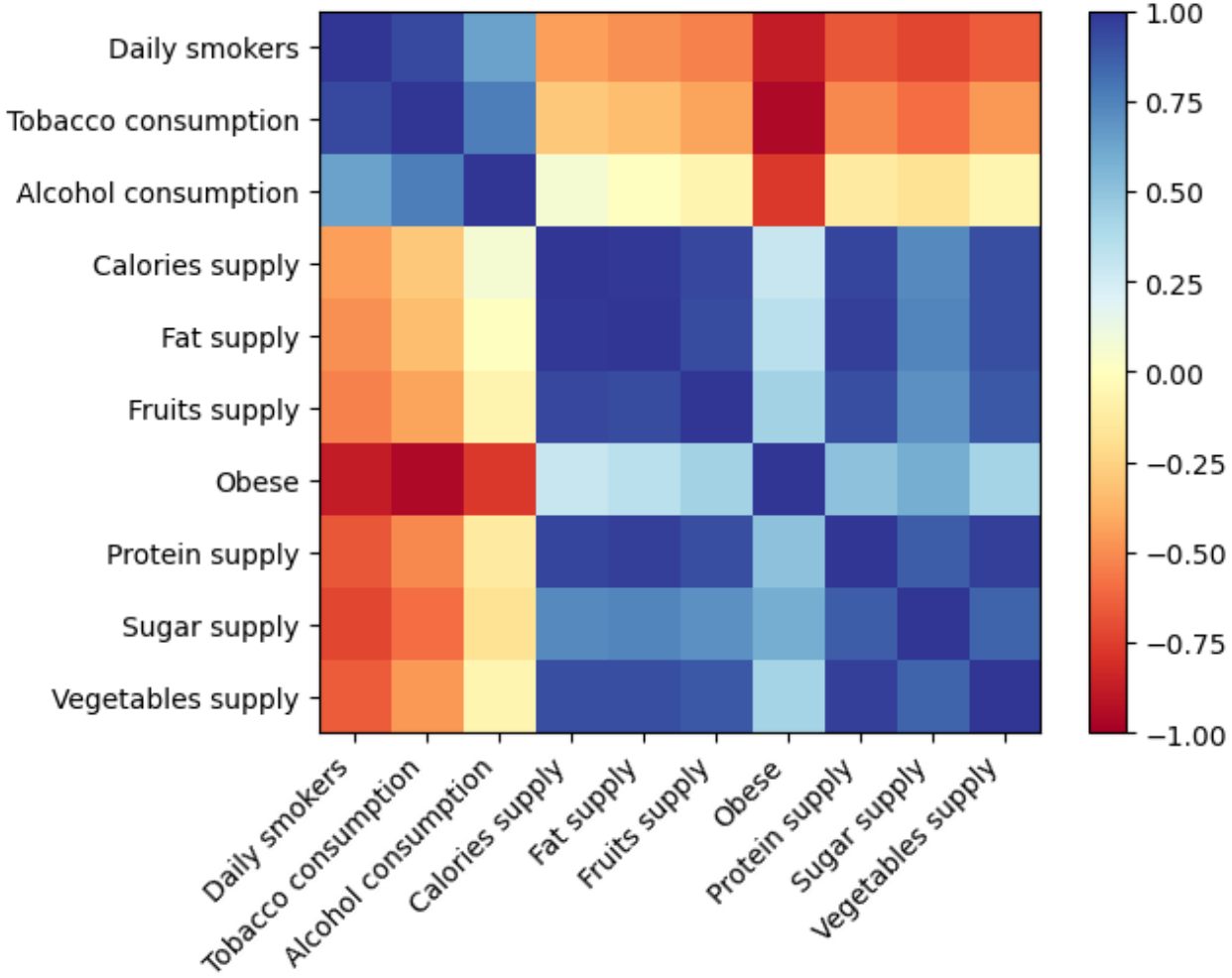
Obese: share of population who are obese.

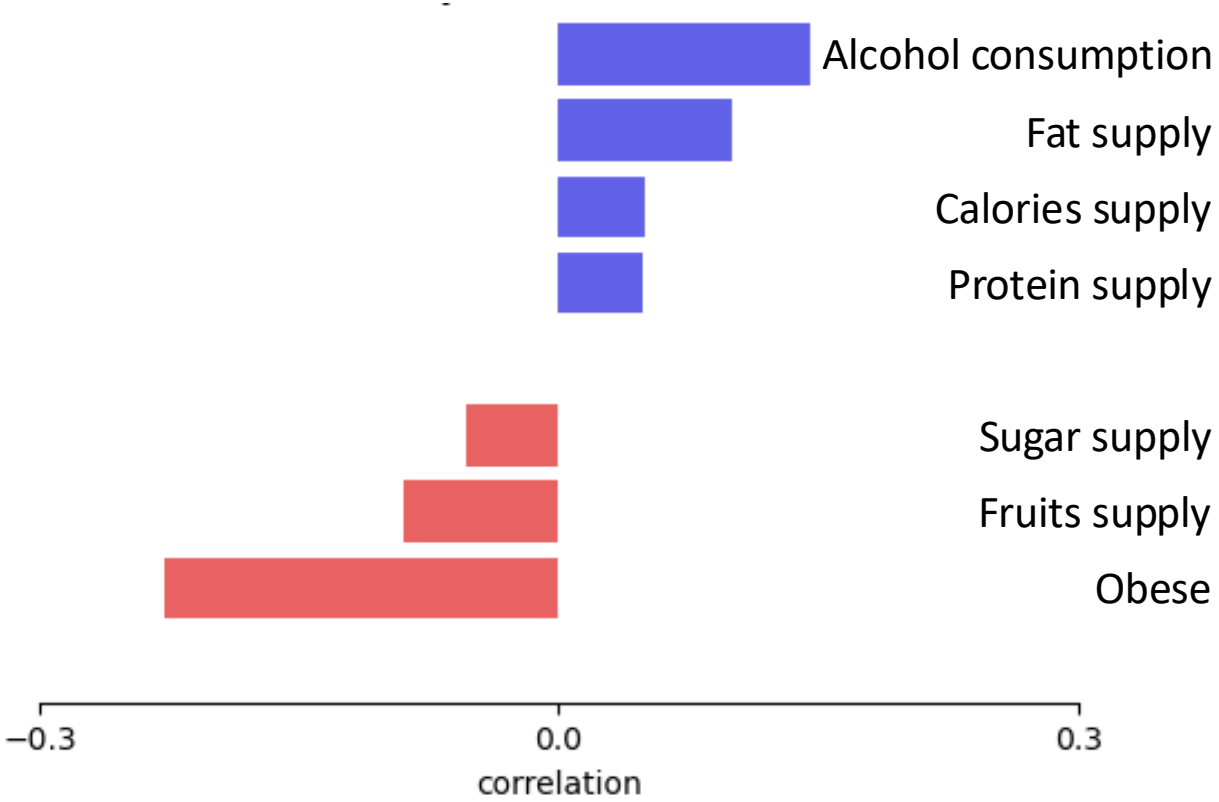
Tobacco, Alcohol consumption: on average per capita.

Calories, Fat, Fruits, Protein, Sugar, Vegetables supply: the amount available per capita.

Source: https://databank.worldbank.org/reports.aspx?source=2&series=SH.ALC.PCAP.FE.LI&country=

Correlation Matrix Between Measures
Across All Countries

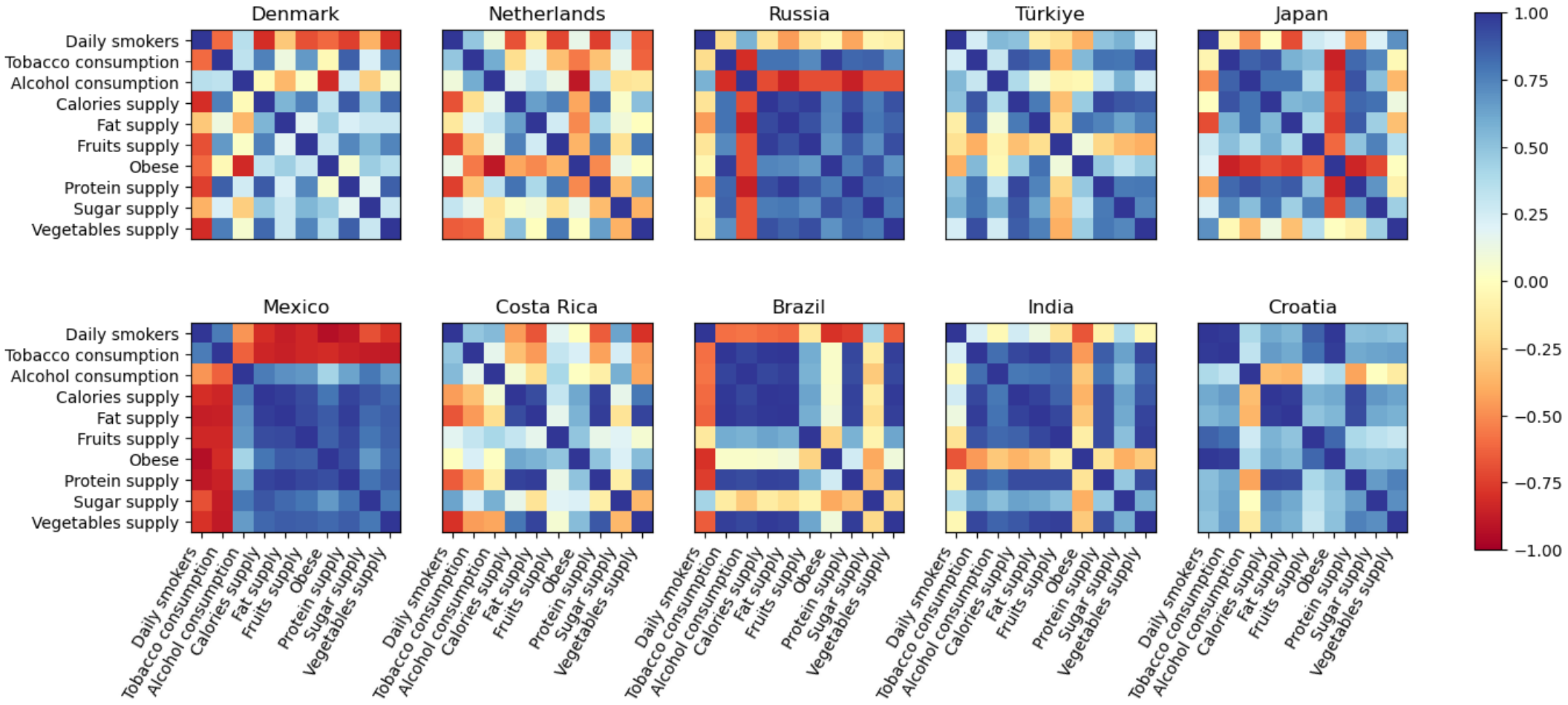




The plot shows a negative correlation globally between *Daily smokers, Tobacco consumption* and the measures *Sugar, Fruits supply* and *the percentage of the population that are Obese*.

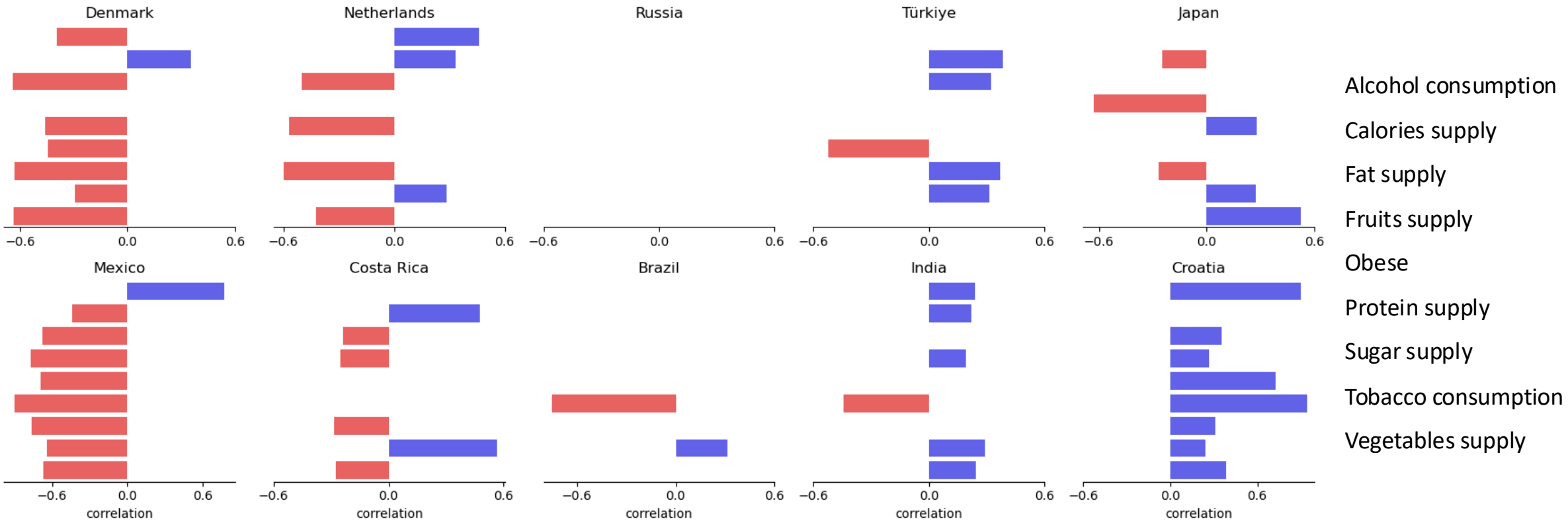
Correlation Matrices of Measures

In the Countries with the Highest (top row) and Lowest (bottom row) Average Percentage of Daily Smokers (PDS)



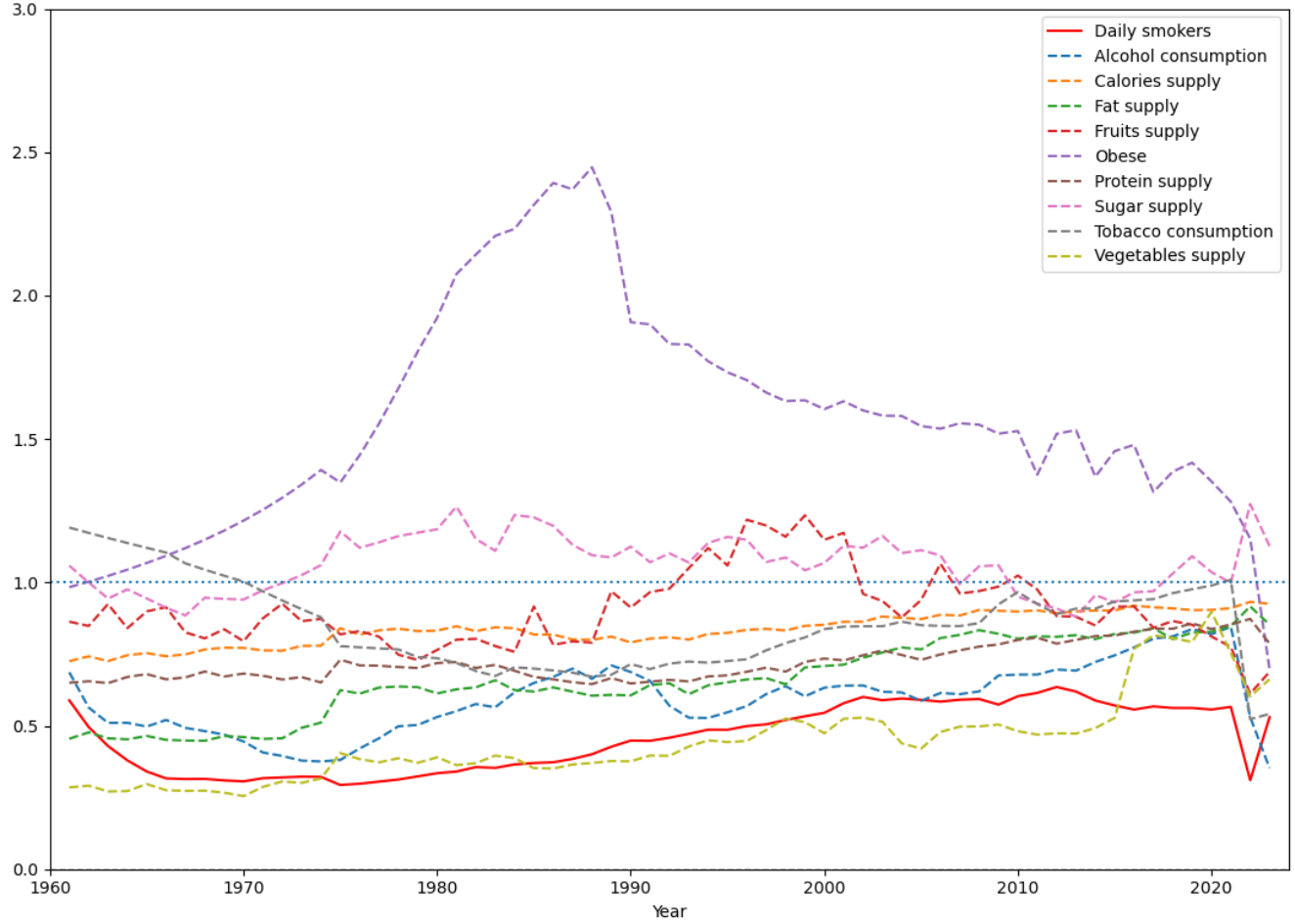
We do not see a clear pattern in the correlation matrices based on their *PDS*.

Correlation of *PDS* with Other Measures

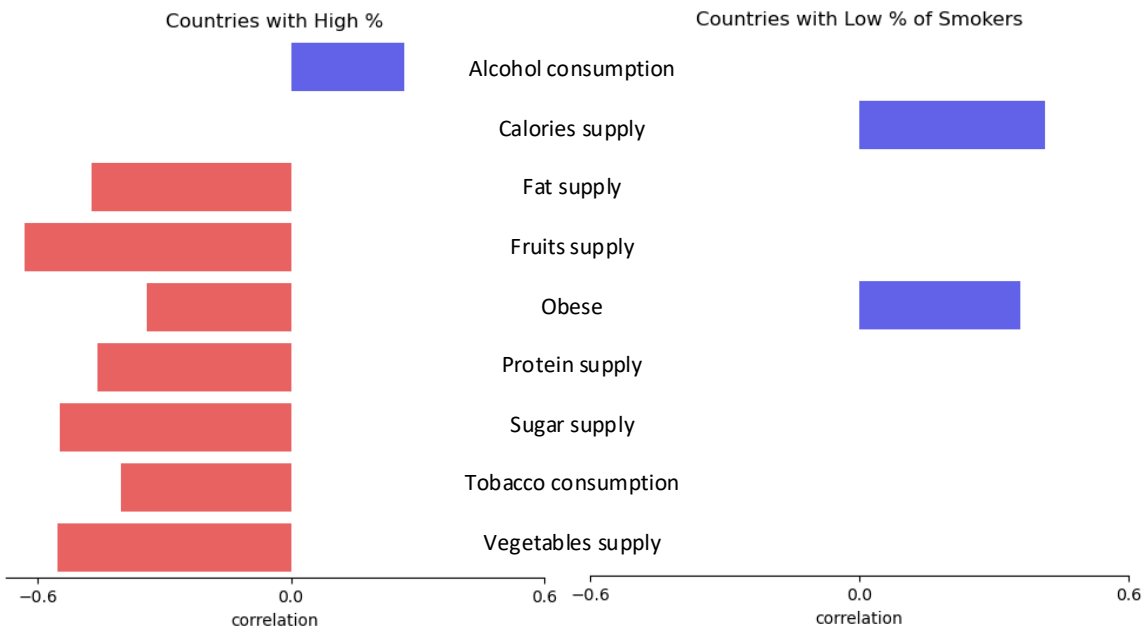


Though we do not see a clear pattern in the correlation coefficients among our two group.

The Ratio of Measures in Countries
with *low vs high PDS*



Correlation of PDS with Other Measures



The measure *Obese* has a higher percentage in countries with a lower *PDS*.

This is consistent with our previous findings.